A STATISTICAL ANALYSIS OF RISK FACTORS IN HEART DISEASE USING MULTIVARIATE ANALYSIS TECHNIQUES

A project work submitted to

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(Affiliated to University of Madras)

In partial fulfilment of the requirements for the award of the degree of

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IN

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Submitted by

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CERTIFICATE

This is to certify that the project work entitled, **"A STATISTICAL ANALYSIS OF RISK FACTORS IN HEART DISEASE USING MULTIVARIATE ANALYSIS TECHNIQUES"** is a bonafide record of research work done by **KARTHIK S (Reg. No: 2113031089003)** a candidate of the **Department of Statistics, Dr. Ambedkar Government Arts College (Autonomous), Vyasarpadi, Chennai-600039,** Under my guidance and supervision and this project work has not previously formed the basis for the award of any Degree and Diploma. It is entirely an independent and original work on the part of the candidate under my supervision.

Place: Chennai **Prof. Dr. V. PRAKASH**
Date: 22/03/2024 **Supervisor and Guide**

Prof. Dr. V. PRAKASH
Head of the Department **External Supervisor**

A STATISTICAL ANALYSIS OF RISK FACTORS IN HEART DISEASE USING MULTIVARIATE ANALYSIS TECHNIQUES

ABSTRACT

Heart disease remains a leading cause of mortality worldwide, encompassing various conditions affecting the heart's structure and function. These abstract reviews the primary risk factors associated with heart disease, including modifiable factors such as hypertension, high cholesterol, smoking, obesity, diabetes, and sedentary lifestyle, as well as non-modifiable factors like age, gender, and family history.

The findings from this study promise to enhance our understanding of the complex dynamics influencing heart disease risk, potentially contributing to more targeted prevention and intervention strategies in public health initiatives.

OBJECTIVE

World Health Organization has estimated 12 million deaths occur worldwide, every year due to Heart diseases. Half the deaths in the India and other developed countries are due to cardio vascular diseases. The early prognosis of cardiovascular diseases can aid in making decisions on lifestyle changes in high risk patients and in turn reduce the complications. This project intends to pinpoint the most relevant/risk factors of heart disease as well as predict the overall risk using multivariate analysis techniques.

AIM

The aim of this project is to employ multivariate statistical analysis techniques to identify and quantify the most influential risk factors associated with heart disease, utilizing a comprehensive dataset.

DATA

It’s a Secondary data collected from the internet via Kaggle.

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CHAPTER-I

1.0 INRODUCTION

1.1 CARDIO VASCULAR DISEASE



Heart Disease :

Heart and blood vessel disease, also called Heart Disease, includes numerous problems, many of which are related to atherosclerosis.

Atherosclerosis is a condition that develops when a substance called plaque builds up in the walls of the arteries. This buildup narrows the arteries, making it harder for blood to flow through. If a blood clot forms, it can block the blood flow. This can cause a heart attack or stroke.

Heart attack :

A heart attack occurs when the blood flow to a part of the heart is blocked by a blood clot. If this clot cuts off the blood flow completely, the part of the heart muscle supplied by that artery begins to die. Most people survive their first heart attack and return to their normal lives, enjoying many more productive years.

Heart failure :

Heart failure, sometimes called congestive heart failure, means the heart isn't pumping blood as well as it should. Heart failure does not mean that the heart stops beating — that's a common misperception. Instead, the heart keeps working, but the body's need for blood and oxygen isn't being met

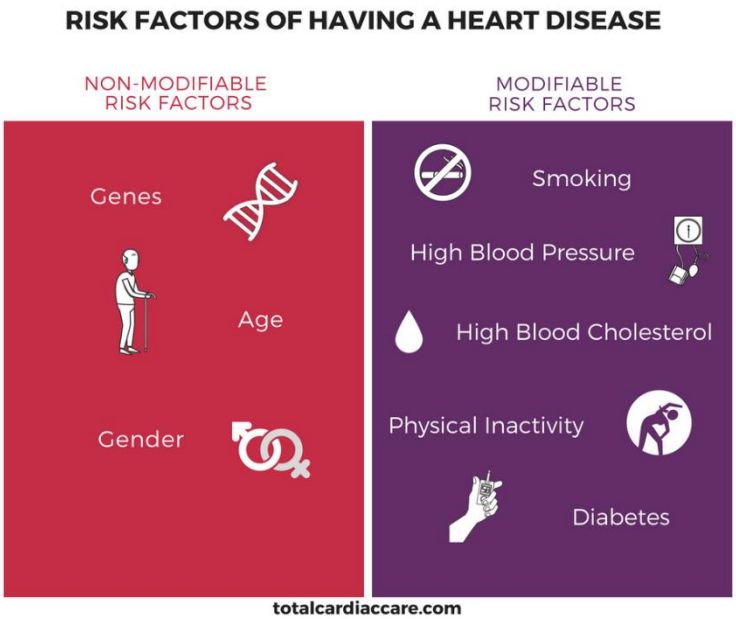
1.2 What are cardiovascular diseases?

Cardiovascular diseases (CVDs) are a group of disorders of the heart and blood vessels. They include:

- coronary heart disease – a disease of the blood vessels supplying the heart muscle;
- cerebrovascular disease – a disease of the blood vessels supplying the brain;
- peripheral arterial disease – a disease of blood vessels supplying the arms and legs;
- rheumatic heart disease – damage to the heart muscle and heart valves from rheumatic fever, caused by streptococcal bacteria;
- congenital heart disease – birth defects that affect the normal development and functioning of the heart caused by malformations of the heart structure from birth; and
- deep vein thrombosis and pulmonary embolism – blood clots in the leg veins, which can dislodge and move to the heart and lungs.

Heart attacks and strokes are usually acute events and are mainly caused by a blockage that prevents blood from flowing to the heart or brain. The most common reason for this is a build-up of fatty deposits on the inner walls of the blood vessels that supply the heart or brain. Strokes can be caused by bleeding from a blood vessel in the brain or from blood clots.

1.3 What are the risk factors for cardiovascular disease?



The most important behavioral risk factors of heart disease and stroke are unhealthy diet, physical inactivity, tobacco use and harmful use of alcohol. The effects of behavioral risk factors may show up in individuals as raised blood pressure, raised blood glucose, raised blood lipids, and overweight and obesity. These “intermediate risks factors” can be measured in primary care facilities and indicate an increased risk of heart attack, stroke, heart failure and other complications.

In addition, drug treatment of hypertension, diabetes and high blood lipids are necessary to reduce cardiovascular risk and prevent heart attacks and strokes among people with these conditions.

1.4 What are common symptoms of cardiovascular diseases?

Symptoms of heart attacks and strokes

Often, there are no symptoms of the underlying disease of the blood vessels. A heart attack or stroke may be the first sign of underlying disease. Symptoms of a heart attack include:

- pain or discomfort in the centre of the chest; and/or
- pain or discomfort in the arms, the left shoulder, elbows, jaw, or back.

In addition the person may experience difficulty in breathing or shortness of breath; nausea or vomiting; light-headedness or faintness; a cold sweat; and turning pale. Women are more likely than men to have shortness of breath, nausea, vomiting, and back or jaw pain.

1.5 Heart attacks are on the rise in patients aged 20-30 years old



Not long ago, heart attacks were primarily a problem faced by older adults. It was rare for anyone younger than 40 to have a heart attack. Now 1 in 5 heart attack patients are younger than 40 years of age.

Here’s another troubling fact to highlight the problem: Having a heart attack in your 20s or early 30s is more common. Between the years 2000-2016, the heart attack rate increased by 2% every year in this young age group.

Your outlook isn’t better following a heart attack just because you’re younger. Patients who have a heart attack in their 20s or 30s face the same risks as older patients. Once you have that first heart attack, you have the same chance of dying from a second major heart event or a stroke regardless of your age.

1.6 Young adults are increasingly diagnosed with hypertension

Hypertension, or high blood pressure, is one of the biggest risk factors for cardiovascular disease. Unfortunately, just like the trend in heart attacks, the incidence of hypertension is rising faster in young adults than in older adults. High blood pressure makes your heart muscles thicken, harms your blood vessels, and increases your risk of a heart attack.

1.7 Overweight and obesity are key players in heart attacks

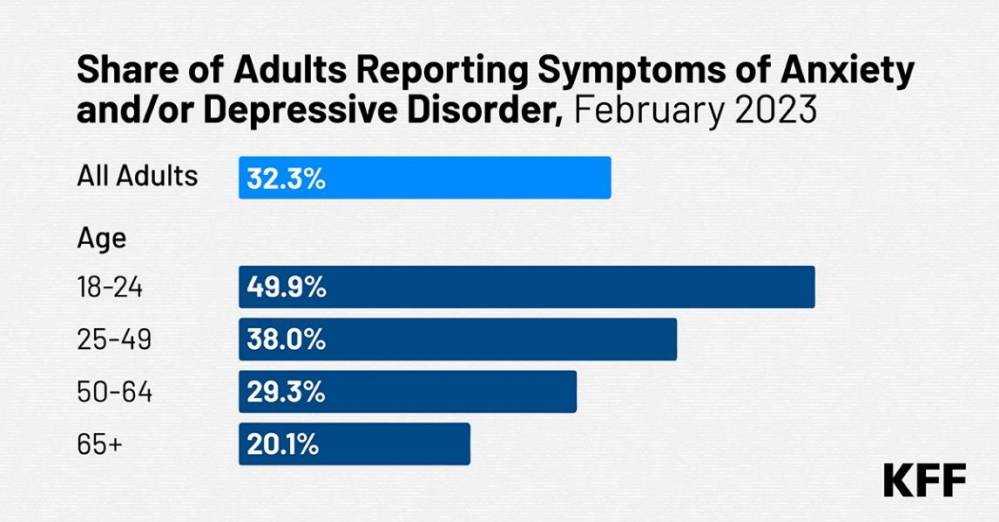
Being overweight has a significant influence on your risk of having a heart attack for several reasons. For starters, carrying extra weight places a high demand on your heart. Obesity alone makes you more likely to have a heart attack even if you’re otherwise healthy.

However, patients who are overweight and obese often have other health conditions that detract from their heart health, such as high blood pressure, diabetes, and high cholesterol.

1.8 Smoking cigarettes and vaping are major risk factors for a heart attack

Of all the things that contribute to heart attacks in young adults, smoking cigarettes is one of the top risk factors. Your risk of a heart attack increases in direct proportion to the number of cigarettes you smoke. Smoking one pack a day more than doubles your risk for a heart attack compared with nonsmokers.

1.9 Rise of depression amongst young adults in India



The United Nations defines youth as the people between the age group of 15 to 24 years. This age group comprises 10% of the Indian population. According to the World Health Organization, depression affects over 300 million people globally, and it is predicted to be the leading cause of disability by 2030. Recent studies indicate that depression rates are increasing among young people in India with prevalence rates ranging from 31% to 57%. UNICEF also reports that one out of seven young people between the ages of 15 to 24 in India always feels depressed or has little interest in doing things. This suggests a significant prevalence of mental health

challenges among this demographic, which can have serious long-term implications for their well-being and overall quality of life.

It is important to continue conducting research on depression in youth in order to better understand the factors and profiles associated with this condition and to develop effective prevention and remedial strategies

1.10 Preventing heart disease involves adopting a healthy lifestyle and managing risk factors. Here are some key steps to prevent heart disease:



Healthy Diet

- Consume a balanced diet rich in fruits, vegetables, whole grains, lean proteins, and healthy fats.
- Limit saturated and trans fats, cholesterol, sodium, and added sugars.
- Choose foods high in fiber, such as oatmeal, beans, fruits, and vegetables.

Regular Exercise

- Aim for at least 150 minutes of moderate-intensity aerobic exercise or 75 minutes of vigorous-intensity exercise per week.
- Incorporate strength training exercises at least two days per week.
- Stay active throughout the day by taking short breaks to walk or stretch if you have a sedentary job.

Maintain a Healthy Weight

- Achieve and maintain a healthy body weight through a combination of diet and exercise.
- Avoid crash diets and focus on gradual, sustainable weight loss.

Quit Smoking

- If you smoke, quit. Smoking significantly increases the risk of heart disease.
- Seek support from healthcare professionals, counseling, or smoking cessation programs

Limit Alcohol Intake

- Drink alcohol in moderation, if at all. Excessive alcohol consumption can contribute to heart disease.

Manage Stress

- Practice stress-reduction techniques such as deep breathing, meditation, yoga, or hobbies.
- Maintain a healthy work-life balance and seek support from friends, family, or professionals if needed.

Get Quality Sleep

- Aim for 7-9 hours of sleep per night.
- Establish a regular sleep schedule and create a conducive sleep environment.

Regular Health Check-ups

- Schedule regular check-ups with your healthcare provider to monitor blood pressure, cholesterol levels, blood sugar levels, and overall heart health.
- Follow your healthcare provider's recommendations for screenings and preventive measures.

Manage Chronic Conditions

- If you have conditions such as high blood pressure, high cholesterol, diabetes, or other chronic diseases, work with your healthcare provider to manage them effectively.
- Take prescribed medications as directed and follow treatment plans.

Know Your Family History

- Be aware of your family's history of heart disease and discuss it with your healthcare provider.
- Understanding your genetic risk factors can help inform your prevention strategies.

By incorporating these lifestyle changes and preventive measures into your daily routine, you can significantly reduce your risk of developing heart disease and improve your overall heart health. Always consult with your healthcare provider before making any significant changes to your lifestyle or starting a new exercise or diet regimen.

CHAPTER-II

2.0 COLLECTION OF DATA

They are two types, of data Primary Data and Secondary Data

- **Primary Data**

Raw data or primary data is a term for data collected at source. This type of information is obtained directly from first hand sources by means of surveys, observations and experimentation and not subjected to any processing or manipulation and also called primary data.

Primary data is collected in the course of doing experimental or descriptive research by doing experiments performing surveys or by observation or direct communication with respondents.

- **Secondary Data**

A researcher can obtain secondary data from various sources. Secondary data may either be published data or unpublished data.

Published data are available in:

- ❖ Publication of government
- ❖ Technical and trade journals
- ❖ Reports of various business, banks etc.
- ❖ Public records
- ❖ Statistical or historical documents.

Unpublished data may be found in letters, diaries, unpublished biographies or work.

2.1 BAR GRAPH

A Bar graph is a pictorial rendition of statistical data in which the independent variable can attain only certain discrete values. The independent variable may be discrete or continuous. The most common form of bar graph is the vertical bar graph, also called a column graph. In a vertical bar graph, values of the independent variable are plotted along a horizontal axis from left to right. Function values are shown as shaded or colored vertical bars of equal thickness extending upward from the horizontal axis to various heights. In a horizontal bar graph, the independent variable is plotted along a vertical axis from the bottom up. Values of function are shown as shaded or colored horizontal bars of equal thickness extending toward the right, with their left ends vertically aligned.

The illustration is a vertical bar graph depicting the result of a test given to hypothetical class of students. Each is denoted by a blue vertical bar of a certain height. The total of the percentage is equal 100. The percentage of rainfall receiving a specific grade is directly proportional to the height of the bar representing that grade. The bar thickness is chosen only for illustration clarity.

2.2 PIE DIAGRAM

A Pie graph is a specialized graph used in statistics. The independent variable is plotted around a circle in either a clockwise direction or a counter clockwise direction. The independent variable is rendered as an arc whose measure is proportional to the magnitude of the quantity. Each arc is depicted by construction of radial lines from its ends to the circle, creating a wedge shaped "slice". The independent variable can attain finite number of discrete values (for example, five). The dependent variable can attain any value from zero to percent.

A pie chart, sometimes called a circle chart, is a way of summarizing a set of nominal data or displaying the different values of a given variable (e.g., percentage distribution). This type of chart is a circle divided into a series of segments. Each segment represents a particular category.

2.3 HISTOGRAM

A histogram is a visual representation of the distribution of quantitative data. The term was first introduced by Karl Pearson. Histograms give a rough sense of the density of the underlying distribution of the data, and often for density estimation: estimating the probability density function of the underlying variable. The total area of a histogram used for probability density is always normalized to 1. If the length of the intervals on the x-axis are all 1, then a histogram is identical to a relative frequency plot.

A histogram, on the other hand, is a graph that shows the distribution of numerical data. It is a type of bar chart that shows the frequency or number of observations within different numerical ranges, called bins. The bins are usually specified as consecutive, non-overlapping intervals of a variable. The histogram provides a visual representation of the distribution of the data, showing the number of observations that fall within each bin. This can be useful for identifying patterns and trends in the data, and for making comparisons between different datasets

.

2.4 CORRELATION

The degree of relationship between the variables is measured through the correlation Analysis. In a Bi- variate distribution we may be interested to find out if there is any correlation or co- variation between the two variables under study. If the changes in one variable affects in the other variable, the variables are said to be correlated. If the two variable deviate in the same direction, i.e., if the there, correlation is said to direct or positive. But if they constantly deviate in the opposite direction, i.e., if increase in one result in corresponding decrease (or increase) in the other, correlation is said to be negative for example, the correlation between

- The heights and weights of a group of persons, and the income and expenditure are positive
- The correlation between Price and demand of a commodity and volume and pressure of a perfect gas is negative

2.5 REGRESSION

Regression analysis is a form of predictive modelling technique which investigates the relationship between a dependent and independent variables. This technique is used for forecasting, time series modelling and finding the casual effect relationship between the variables.

THE REGRESSION TECHNIQUE IS PRIMARY USED TO:

- ❖ Estimate the relationship that exists, on the average, between the dependent variable and the explanatory variable.
- ❖ Determine the effect of each of the explanatory variables on the dependent variable, controlling the effects of all other explanatory variable
- ❖ Predict the value of dependent variable for a given value of the explanatory variable.

ASSUMPTION OF THE LINEAR REGRESSION MODEL

- Linear functional form
- Fixed independent variables
- Independent observations
- Representative sample and proper specification of the model
- Normality of the residuals or errors
- Equality of the variance of the errors(homogeneity of residual variable)
- No autocorrelation of the errors
- No outlier distortion.

2.6 MULTIPLE LINEAR REGRESSION

The general purpose of multiple regression is to learn more about the relationship between several independent or predictor variables and a dependent or criterion variable. In the social and natural sciences research multiple regression procedures are very widely used. For example, educational researchers might want to learn the best predictors of success in college. Biologists may want to determine the best predictor for the survival of the fish in polluted water. Psychologists may want to determine which personality variable best predicts social adjustment.

$$Y_i = a + bX_i$$

In this relationship, Y and X represent the dependent and independent variables respectively, b is the regression coefficient in the population, a (Y intercept) is the value of Y when X is zero.

In many situations, however, Y may be considered dependent upon more than one variable. Then,

$$Y_j = a + b_iX_{ij}+b_2X_{2j}$$

Here, one variable (Y) is linearly dependent upon a second variable (X_1) and Y is also linearly dependent upon a third variable (X_2). In this particular multiple regression model, we have one dependent variable and two independent variables. The two population parameters b_1, and b_2, are termed as partial regression coefficients; b_1 expresses how much Y would change for a unit change in X_1 if X_2 was held constant. It is sometimes said that b, is a measure of the relationship of Y to X_1 after removing the effect of X_2. The Y intercept, a (sometimes designated as b_0) is the value of Y when both X_1 and X_2, are zero.

If we sample a population containing the three variables (Y, X_1 and X_2) the multiple regression equation may be expressed as

$$Y_j = a +b_1X_{1j}+b_2X_{2j}$$

where Y_J = dependent variable, a is a constant called Y intercept where the value of independent variable X_1 and X_2 is 0, b_2 slope of X_1 with X_2 being held constant, b = slope of X_2 with X_1 being held constant.

2.7 CHI-SQUARE

The χ^{2*} test (pronounced as chi-square test) is one of simplest and most widely used non-parametric tests in statistical work. The symbol χ^2 is the Greek letter Chi. The χ^2 test was first used by Karl Pearson** in the year 1990. The quantity χ^2 describes the magnitude of the discrepancy between theory and observation. It is defined as:

$$X^2=\sum(O-E)^2/E$$

Where O refers to the observed frequencies and E refers to the expected frequencies.

Steps To determine the value of χ^2, the steps required are:

(i) Calculate the expected frequencies. In general the expected frequencies for any all can be calculated from the following equation:

$$E=RT*CT/N$$

E=Expected frequency

RT= the row total for the row containing the cell

CT= the column total for the column containing the cell

N= the total number of observations.

(ii) Take the difference between observed and expected frequencies and obtain the squares of these difference, i.e., obtain the values of $(0-E)^2$.

(iii) Divide the values of $(0-E)^2$ obtained in step (ii) by the respective expected frequency and obtain the total $\sum[(0-E)^2/E]$. This gives the value of χ^2 which can range from zero to infinity. If χ^2 is zero it observed and expected frequencies completely coincide. The greater the discrepancy between the observed and expected frequencies, the greater shall be the value of χ^2.

The calculated value of χ^2 is compared with the table value* of χ^2 for given degrees of freedom at a certain specified level of significance. If at the started level (generally 5% level is selected), the calculated value of χ^2 is more than the table value of χ^2 , the difference between theory and observation is considered to be significant.

IN SIMPLE WORDS

A chi-squared test is a statistical hypothesis test used in the analysis of contingency tables when the sample sizes are large. In simpler terms, this test is primarily used to examine whether two categorical variables are independent in influencing the test statistic.

LEVEL OF SIGNIFICANCE

The probability of rejecting the null hypothesis in a statistical test when it is true called also significance level.

DEGREES OF FREEDOM

Degree of freedom is calculated by using the following formula:

$$DF=(r-1)\ (c-1)$$

Where,

DF=Degree of freedom

r=number of rows

c=number of columns

HYPOTHESIS

Null Hypothesis:

Assumes that there is no association between the two variables

Alternative Hypothesis:

Assumes that there is association between the two variables

HYPOTHESIS TESTING

It is the same for the correlation test for the difference as it for other tests like Non-parametric randomness test, one sample t test, etc. If the calculated value for correlation test is less than the table value, we will accept the null hypothesis. If the calculated value is greater than we will accept the reject the null hypothesis.

P- VALUE

The p- value is the level of marginal significance within a statistical hypothesis test representing the probability of the occurrence of a given event. The P- value used as an alternative to rejection points to provide the smallest level of significance at which the null hypothesis would be rejected. A smaller p- value means that there is stronger evidence in favor of the alternative hypothesis.

P-values are calculated using p- value tables or spreadsheet statistical software. Because different researchers use different levels of significance when examining a question, a reader may sometimes have difficulty comparing results from two different tests. For example, if two studies of returns from two particular assets were undertaken using two different significance levels, a reader could not compare the probability of returns for the two assets easily. For ease of comparison researches often feature the p-value in the hypothesis test and allow the reader to interpret the statistical significance themselves. This is called a p-value approach to hypothesis testing.

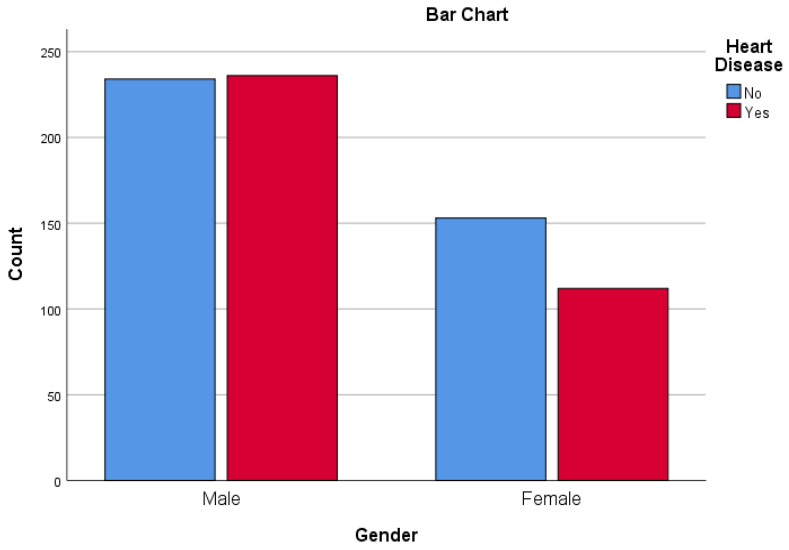
The P value is defined as the probability under the assumption of no effect or no difference (null hypothesis), of obtaining a result equal to or more extreme than what was actually observed. The P stands for probability and measures how likely it is that any observed difference between groups is due to chance.

CHAPTER-III

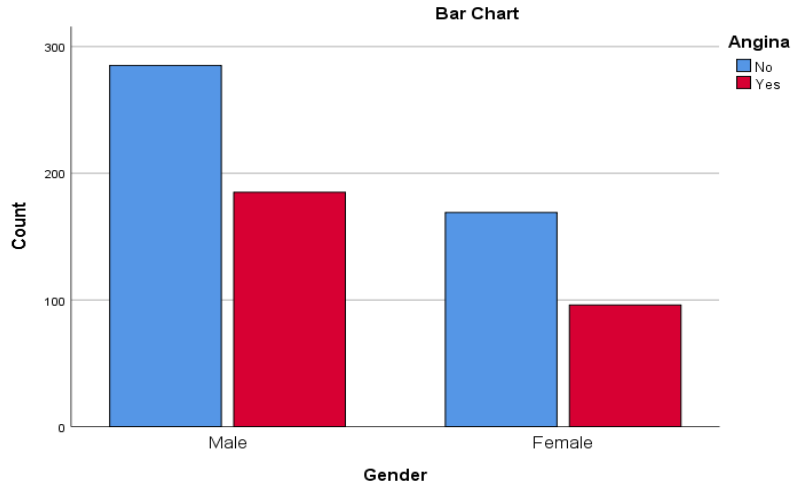
3.0 Graphical Presentation

3.1 BAR CHART

Gender and Heart Disease

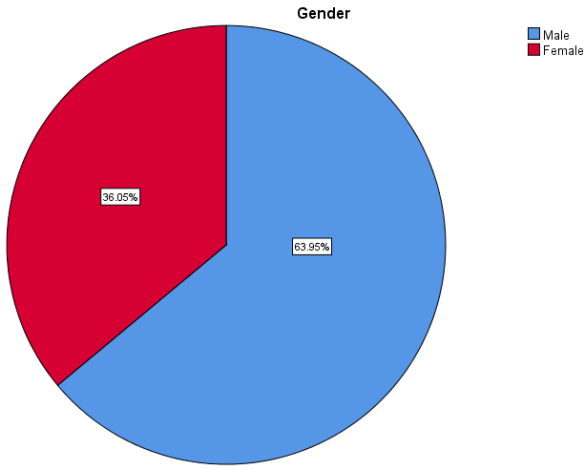


Gender and Angina(Chest Pain)

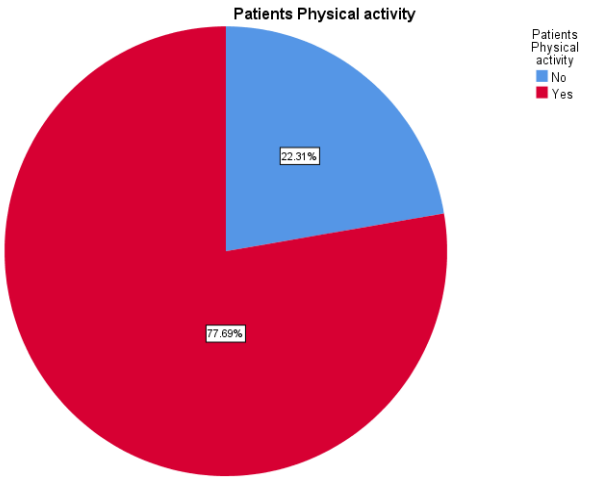


3.2 PIE CHART

GENDER:

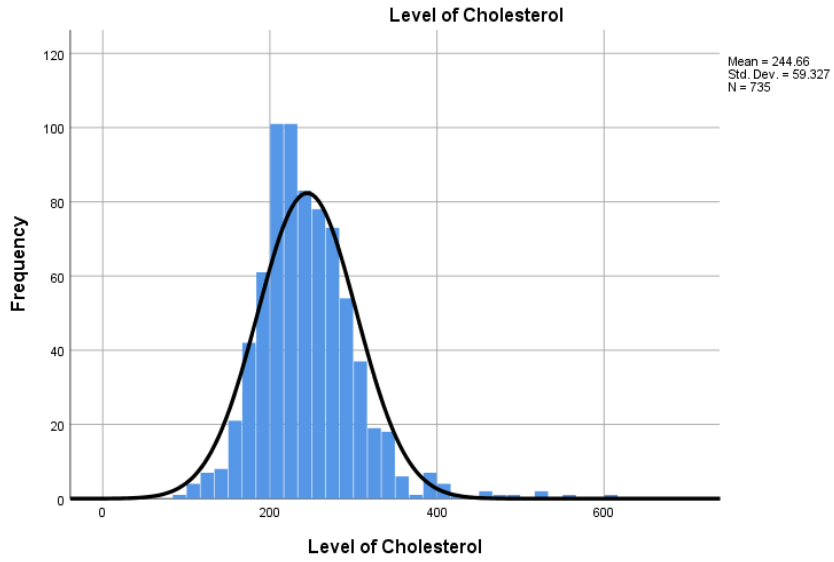


HEART DISEASE:

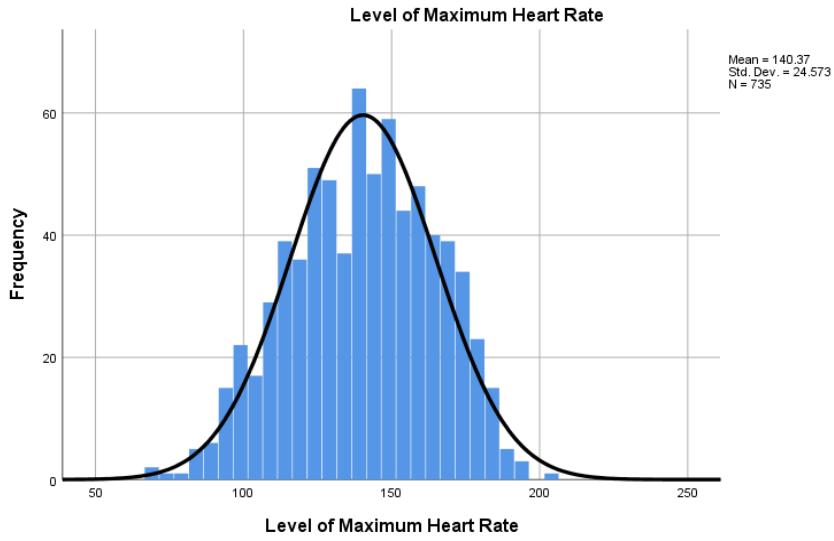


3.3 HISTOGRAM

Level of Cholesterol



Level of Maximum Heart rate



CHAPTER-IV

4.0 ANALYSIS

4.1 DESCRIPTIVE STATISTICS

Descriptive Statistics						
	N	Minimum	Maximum	Mean	Std. Deviation	Variance
Age	735	39	65	53.25	6.981	48.737
Level of Cholesterol	735	85	603	244.66	59.327	3519.652
Level of BMI	735	17	95	27.66	5.662	32.057
Level of Systolic pressure	735	14	200	127.64	17.460	304.846
Level of Maximum Heart Rate	735	69	202	140.37	24.573	603.849
Valid N (listwise)	735					

The Min and Max Values of Variables(Age, Cholesterol, BMI, S_BP, Max_HR) Given Above Table, and also we given Mean, Standard Deviation, Variance of Variables.

4.2 FREQUENCIES OF DATA:

Statistics						
		Level of Cholesterol	Age	Level of BMI	Level of Maximum Heart Rate	Level of Systolic pressure
N	Valid	735	735	735	735	735
	Missing	0	0	0	0	0
Mean		244.66	53.25	27.66	140.37	127.64
Median		237.00	53.89	26.64	140.00	120.00
Mode		254	48[a]	24[a]	150	120
a. Multiple modes exist. The smallest value is shown						

The above Frequency table shown there is no ‘Missing’ values in Data and also we show Centrel Tendency of Scale variables in above table.

4.3 CORRELATION:

We find out the Correlation coefficient for the variables, Age (Years) and Systolic Blood Pressure (mmHg) in Patients Data.

Hypothesis:

Null hypothesis (H_0) : There is no correlation between Age and Systolic blood pressure.

Alternate hypothesis (H_1) : There is correlation between Age and Systolic blood pressure.

OUTPUT:

Correlations

		Age	Level of Systolic pressure
Age	**Pearson Correlation**	**1**	**.188****
	Sig. (2-tailed)		**.000**
	N	**735**	**735**
Level of Systolic pressure	**Pearson Correlation**	**.188****	**1**
	Sig. (2-tailed)	**.000**	
	N	**735**	**735**

****. Correlation is significant at the 0.01 level (2-tailed).**

The Value Against Pearson correlation, i.e., 0.188 is the r-value.

Since, the r-value is positive and significance (2-tailed) value of 0.002 (the p-value) is Below 0.01, we reject the H_0 and Accept the alternate hypothesis and infer that an increase in age really increase the systolic blood pressure (or) Patient's age and Blood pressure are Positively Correlated.

4.4 CHI-SQUARE TEST

The Chi-square test is a statistical test used to determine whether there is a significant association between two categorical variables. It's commonly used in various fields such as medicine, social sciences, and business to analyze data and make inferences about relationships between categorical variables.

Here we test Chi-Square two Categorical variables Exercise Angina and Heart Disease.

Hypothesis:

Null hypothesis (H_0) : There is no significant association between Exercise Angina and Heart Disease.

Alternate hypothesis (H_1) : There is significant association between Exercise Angina and Heart Disease.

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	221.750[a]	1	<.001		
Continuity Correction[b]	219.492	1	<.001		
Likelihood Ratio	235.552	1	<.001		
Fisher's Exact Test				<.001	<.001
Linear-by-Linear Association	221.448	1	<.001		
N of Valid Cases	735				

a. 0 cells (.0%) have expected count less than 5. The minimum expected count is 133.04.

b. Computed only for a 2x2 table

Since, p-value < 0.01, we reject the null hypothesis and we can conclude that there is significance association between Exercise Angina and Heart Disease.

4.5 MULTIPLE REGRESTION

ANOVA[a]						
Model		**Sum of Squares**	**df**	**Mean Square**	**F**	**Sig.**
1	**Regression**	**12555.189**	**4**	**3138.797**	**10.849**	**.000[b]**
	Residual	**211201.986**	**730**	**289.318**		
	Total	**223757.176**	**734**			
a. Dependent Variable: Level of Systolic blood pressure						
b. Predictors: (Constant), Level of Cholesterol, Age, Level of Maximum Heart Rate, Level of BMI						

Based on the ANOVA results for predicting Systolic blood pressure using predictors such as level of Cholesterol, Age, Level of Max Heart rate and BMI it seems that the overall model is significant ($p < .001$), indicating that at least one of the predictors significantly contributes to the prediction of Systolic Blood Pressure. Let's break down the significant predictors: level of Cholesterol, Age, Level of Max Heart rate and BMI collectively contribute significantly to the prediction of Systolic Blood Pressure, as indicated by the significant F-statistic ($F = 10.849$, $p < .001$).

This suggests that at least one of the predictors (level of Cholesterol, Age, Level of Max Heart rate and BMI) has a significant relationship with Systolic Blood Pressure. To further interpret the specific contributions of each predictor, you may need to examine the coefficients and their associated p-values in the regression output. These coefficients indicate the direction and strength of the relationship between each predictor and the outcome variable (Level of Systolic Blood Pressure).

Model Summary[b]									
					Change Statistics				
Model	**R**	**R Square**	**Adjusted R Square**	**Std. Error of the Estimate**	**R Square Change**	**F Change**	**df1**	**df2**	**Sig. F Change**
1	**.237[a]**	**.056**	**.051**	**17.009**	**.056**	**10.849**	**4**	**730**	**.000**
a. Predictors: (Constant), Level of Cholesterol, Age, Level of Maximum Heart Rate, Level of BMI									
b. Dependent Variable: Level of Systolic blood pressure.									

R Square: The coefficient of determination (R Square) represents the proportion of variance in the dependent variable (Systolic Blood Pressure) that is explained by the independent variables (level of Cholesterol, Age, Level of Max Heart rate and BMI). Here, R Square = 0.51, indicating

that approximately 51% of the variance in Level of Systolic Blood Pressure is accounted for by the predictors.

Overall, the model appears to have moderate predictive power, with the predictors collectively explaining approximately 51% of the variance in Level of Systolic Blood Pressure. The significant F-statistic ($p < .001$) indicates that the overall model is statistically significant.

Here are the coefficients for the predictors in the regression model predicting Level of Systolic Blood Pressure:

Coefficients[a]						
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	92.574	7.008		13.209	.000
	Age	.449	.090	.180	4.980	.000
	Level of BMI	.444	.111	.144	3.991	.000
	Level of Maximum Heart Rate	.005	.026	.006	.177	.860
	Level of Cholesterol	-.007	.011	-.025	-.682	.496
a. Dependent Variable: Level of Systolic blood pressure						

Unstandardized Coefficients (B): These coefficients represent the change in the dependent variable (Systolic Blood Pressure) for a one-unit change in the predictor, holding all other predictors constant.

Standard Error: This indicates the variability of the coefficient estimate. Lower values suggest greater precision.

Sig.: This column represents the significance level (p-value) associated with each predictor. It indicates whether each predictor has a statistically significant relationship with the outcome variable Systolic Blood Pressure.

Results Based on the coefficients:

Since, Age and Level of BMI have statistically significant relationships with Level of Systolic Blood Pressure ($p < 0.01$).

Level of Cholesterol and Level of Max Heart Rate do not have statistically significant relationships with Level of Systolic Blood pressure ($p > .05$) in this model.

CHAPTER-V

5.0 CONCLUSION

Our project aimed to predict and analyze risk factors for heart disease. Given that our dependent variable, heart disease, is categorical, we opted to explore associations between continuous variables using correlation analysis. Additionally, we employed the chi-square test for associations between two categorical variables. Further, we conducted multiple regression analysis to examine the relationship between Systolic blood pressure, a continuous variable.

To proceed with our analysis, we have to use binary logistic regression due to the categorical nature of our dependent variable (Binary codes).

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